

KSHITIJ DURAPHE

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EDUCATION

Boston University

Boston, MA

M.S., Electrical and Computer Engineering (with Thesis)

Sep 2022 – May 2024

Advisor: Prof. Joshua Semeter

Degree GPA: 3.8/4.0

Selected coursework: Cosmic Plasma Physics · Fourier Optics · Quantum Mechanics and Semiconductor Physics · Quantum Computing · Signal Processing · Machine Learning · Advanced Algorithms · Advanced Discrete Mathematics

College of Engineering Pune

Pune, India

B.Tech., Electrical Engineering (Minor: Computer Science)

Aug 2018 – June 2022

Advisors: Prof. Archana Thosar, Prof. Suhas Kakade

GPA: 9.11/10.0 (Department Bronze Medalist)

Selected coursework: Probability Theory and Statistical Inference · Optics and Modern Physics · Electromagnetism · Numerical Methods · Vector Calculus · Partial Differential Equations · Ordinary Differential Equations · Complex Analysis

RESEARCH INTERESTS

High-Energy Astrophysics; Foundation Models; Mechanistic Interpretability; Space Physics; Time-Domain Astrophysics.

PUBLICATIONS

1. Shared Geometry Is Not Shared Physics: A Layerwise Test of the Platonic Representation Hypothesis in Astronomy

K. Duraphe, A. Kannappan, D. L. Chan, O. Famutimi, Y. Ejlagiri, M. J. Smith, J. F. Wu
NeurIPS Interp4Discovery Workshop, 2026

2. State-Dependent X-ray Variability in Cygnus X-1: A 12-Year NuSTAR Timing Study of Accretion Flow Geometry

K. Duraphe, G. Bhatta, K. Mandar, C. Khanal, et al.
The Astrophysical Journal, 2026

3. A NuSTAR Reflection-Spectroscopy Survey of Cygnus X-1

K. Duraphe, G. Bhatta, et al.
Submitted to The Astrophysical Journal, 2026

4. **What AstroPT knows about galaxies, and what that can teach us about LLMs**
K. Duraphe, A. Kumar, S. Sourav, M. J. Smith
 COLM Sci-FM Workshop, 2026
5. **Smartphone Carrier Phase TEC: A Study Across Ionospheric Spatio-Temporal Scales**
 N. Servan-Schreiber, J. Semeter, **K. Duraphe**, et al.
 Space Weather, 2026
6. **The Platonic Universe: Do Foundation Models See the Same Sky?**
v1: **K. Duraphe**, M. J. Smith, J. F. Wu, S. Sourav (*co-first author*)
 NeurIPS ML4PS Workshop, 2025 — [Spotlight \(top 1%\)](#)
v2: T. Borrell, S. Dillmann, **K. Duraphe**, F. Eris, K. Iyer, A. Khederlarian, A. Kumar, G. Marraffini, M. J. Smith, S. Sourav, R. Di Tella, J. F. Wu (submitted to ICLR 2027 main track)
7. **Optimizing Solar Panel Tilt using Machine Learning Techniques**
K. Duraphe, S. Kakade, et al.
 GPECOM, 2021

THESES

1. **Data-Driven Techniques to Advance Our Understanding of the STEVE Phenomenon**
Kshitij Duraphe, Joshua Semeter
 M.S. Thesis, Department of Electrical and Computer Engineering, Boston University, 2024
2. **Design of an Automated Radio Telescope for Observing the 21 cm Hydrogen Line**
Kshitij Duraphe, Archana Thosar
 B.Tech. Thesis, Department of Electrical Engineering, College of Engineering Pune, 2022

RESEARCH EXPERIENCE

University of Zielona Góra

Zielona Góra, Poland (Remote)

Research Assistant

Mar 2025 – Present

Advisor: Prof. **Gopal Bhatta**

- First-authored a 12-year NuSTAR archival timing study of Cygnus X-1 (*The Astrophysical Journal*, 2026), mapping accretion-flow geometry across spectral states via hard X-ray spectral-timing analysis.
- Identified a previously unrecognized failed state transition and characterized state-dependent variability (power spectra, rms-flux relations, lag-frequency spectra), placing direct constraints on the corona-disk geometry near the black hole.
- Reduced and analyzed the complete public NuSTAR Cyg X-1 archive with HEASoft / NuSTARDAS / Stingray / AstroPy; led manuscript writing and referee response.
- Currently leading a follow-up multi-epoch **spectral** study of Cyg X-1 with submitted to *The Astrophysical Journal* ([link](#)), characterizing coronal and reflection properties across spectral states and investigating the potential failed state transition.

UniverseTBD Collaboration

Remote

Collaborators: Dr. **Michael J. Smith**, Dr. **John F. Wu**

- First author on *The Platonic Universe* (**Spotlight, top 1%** at NeurIPS ML4PS 2025; expanded 12-author version submitted to ICLR 2027 main track). Designed the evaluation framework comparing foundation-model representations across eleven model families from 10M to 10B parameters on JWST, HSC, and Legacy Survey imaging and DESI spectroscopy under model and data scaling; showed that physics performance tracks *local* embedding geometry (mutual k -NN) but not *global* similarity (CKA), motivating an *Aristotelian* rather than Platonic Representation Hypothesis.
- First author on *What AstroPT knows about galaxies, and what that can teach us about LLMs* (Sci-FM Workshop, COLM 2026). Probed frozen AstroPT representations across checkpoints, depths, model sizes, and training objectives; found that galaxy properties emerge in a fixed order tracking their known physical difficulty — pixel-measurable properties early and shallow, inferred properties later and deeper — with the ordering stable across objectives and scales.
- First author on *Shared Geometry Is Not Shared Physics: A Layerwise Test of the Platonic Representation Hypothesis in Astronomy* (NeurIPS Interpretability for Discovery Workshop, 2026), separating layerwise where representations agree geometrically from where they agree physically. **Mentored 4 junior researchers** as part of **EleutherAI**'s SOAR 2026 program.
- Earlier extension of the Platonic Universe work accepted to the ICML 2026 Mechanistic Interpretability Workshop, where I served as an Outstanding Reviewer.
- Creator of the open-source **platonic-universe** package (Python / PyTorch): data loaders, evaluation harness, reproducible configs, test infrastructure.
- Invited **AstroAI talk at the Harvard Center for Astrophysics** (Jan 2026) and **NeurIPS ML4PS oral presentation** (Dec 2025). Spotlight award (top 1% paper) NeurIPS ML4PS 2025.
- Used NCSA DeltaAI compute allocation **PHY250286** — *The Platonic Universe: Do Foundation Models See the Same Sky?* — 6,000 GPU-hours (PI: M. J. Smith; Sep 2025 – Sep 2026); actively running training and evaluation experiments under the allocation.

Space Physics Lab, Boston University

Boston, MA

Graduate Research Assistant

Oct 2022 – May 2024

Advisor: Prof. Joshua Semeter

MS Thesis: Data-Driven Techniques for STEVE

Jul 2023 – May 2024

- Characterized **STEVE** morphology and kinetics from high-resolution citizen-science imagery; developed computer-vision tracking methods that quantified distinct westward velocities for columnar (13 km/s) and picket-fence (1 km/s) features.
- Developed and contributed to various open-source Python and C++ libraries for high-cadence GNSS carrier-phase processing and Total Electron Content (TEC) analysis; identified a candidate two-stage TEC signature of STEVE passage distinct from typical substorms.
- Developed automated STEVE detection for noisy All-Sky Imager data by adapting faint-source astronomical algorithms and evaluating deep-learning models (ConvNeXt), with modular superresolution, inpainting, and time-series classification components.

NASA Life on Mars Initiative - BU Center for Computing and Data Sciences

Jul 2023 – Jan 2024

- Developed GPS-signal propagation models and spatiotemporal interpolation of sparse Martian-ionosphere TEC data (classical and deep-learning baselines) from MARSIS.

MS Project — Plasma Dynamics of STEVE

Oct 2022 – May 2023

- Ran 3D forward-modeling simulations of the ionospheric plasma regime conducive to STEVE formation using the open-source GEMINI3D fluid-electrodynamic model; analyzed density structures, temperature enhancements, and flow shears driven by STEVE-like conditions.

IIT Bombay

Mumbai, India

Research Intern — Bayesian Binary-Star Analysis

May 2021 – Aug 2021

- Performed Bayesian analysis of the eclipsing binary **QX Cas** using PHOEBE and MCMC; developed a custom 3-body dynamics simulator with a differential-evolution solver to constrain parameters of a hypothesized tertiary companion.

College of Engineering Pune

Pune, India

Undergraduate Research Assistant

Jan 2021 – May 2022

Advisors: Prof. Archana Thosar, Prof. Suhas Kakade

Signal Processing Lab — Data-Driven Solar Panel Tilt Optimization

Jan 2021 – Apr 2021

Advisor: Prof. Suhas Kakade

- Formulated optimal solar-panel tilt as a regression problem over multi-year irradiance and meteorological time-series; benchmarked gradient-boosted, kernel-based, and shallow-network estimators against the standard latitude-based heuristic.
- Showed measurable energy-yield gains across heterogeneous Indian and American climatological zones; first-authored the resulting publication at **IEEE GPECOM 2021**.

B.Tech. Thesis: 21 cm Radio Telescope

Aug 2021 – May 2022

- Designed and built a high-gain (20 dB) pyramidal horn radio telescope for 21 cm hydrogen-line observations (Ansys HFSS-simulated waveguide/feed).
- Integrated RTL-SDR + LNA front-end with **Python and C++** WOLA-FFT spectral-analysis software (**VIRGO** spectrometer-based); recovered a partial galactic rotation curve from 21 cm emission.

Naxxatra Equinox Initiative

Bangalore, India

Research Intern — Stellar Structure and Compact Objects

Jul 2020 – Oct 2020

- Derived the equations of stellar structure incorporating hydrostatic equilibrium and relativistic electron-degeneracy pressure; numerically solved the coupled ODEs with a 4th-order Runge-Kutta scheme in Python to compute the white-dwarf mass-radius relation and determine the Chandrasekhar limit.

SELECTED TALKS AND POSTERS

- *The Platonic Universe: Do Foundation Models See the Same Sky?* (invited, long version) **AstroAI, Harvard CfA, Jan 2026**
- *The Platonic Universe: Do Foundation Models See the Same Sky?* (oral) **NeurIPS ML4PS, Dec 2025**
- *Using high-rate dual-frequency cellphones to study the April 8th solar eclipse* **CEDAR Workshop, Jun 2024**

AWARDS AND HONORS

- NeurIPS ML4PS 2025 — **Spotlight Paper** (top 1%) 2025
- MS Ambassador, Boston University ECE Department 2024
- 2nd / 100+ international teams, MIT iQuHack 2023 (IBM x Covalent Challenge) 2023
- Bronze Medalist, Electrical Engineering Department, College of Engineering Pune 2022

TEACHING, MENTORING, AND OPEN-SOURCE COMMUNITY

Computational Research Access NEtwork (CRANE) Physics

Winter 2024, 2025

TA / Mentor

- Topics: signal processing, Python, numerical methods, machine learning, PIC simulations. Wrote tutorial materials and supported students during hands-on sessions across both cohorts.

ENG EC520 Image Processing (Graduate Course), Boston University

Spring 2023

Grader

Instructor: Prof. **Janusz Konrad**

- Graded problem sets and projects on filtering, compression, reconstruction, and computer-vision algorithms for a graduate-level image-processing course.

ArkOS — MIT Student Information Processing Board (SIPB)

Cambridge, MA

Contributor — DevOps and Documentation

2025 – Present

- Lead DevOps and documentation for an open-source local-LLM agent platform; maintain GitHub Actions CI/CD, contributor onboarding, and developer-facing documentation.

rinexpy — GNSS/RINEX processing library

Boston, MA

Author and Maintainer

2023 – Present

- Python library for fast RINEX reading and access, written for high-cadence GNSS carrier-phase work and released as a superset of **georinex**: merging and header/observation-type editing, automatic clock-jump detection and correction, higher-order ionospheric corrections, BINEX and RTCM conversion, and an interactive web-based file viewer.

COEP Astronomy Club

Pune, India

Head of Projects

2018 – 2022

- Mentored 30 undergraduates on processing astronomical data, operating telescopes, and building reflector optics.
- Ran outreach sessions at schools for autistic children on identifying constellations in the night sky.

KEY SKILLS

Astronomy tooling: AstroPy, HEASoft, NuSTARDAS, Stingray, PHOEBE, GEMINI3D, FITS / HDF5 I/O

Scientific Python ecosystem: NumPy, SciPy, Matplotlib, pandas, PyTorch, scikit-learn

Languages: Python (primary), C, C++, SQL, MATLAB

Statistical and numerical methods: Bayesian inference (MCMC), spectral-timing analysis, time-series analysis, ODE/PDE numerical solvers, Runge–Kutta

Software engineering: Git / GitHub, GitHub Actions CI/CD, pytest, Sphinx-style documentation, open-source contribution and code review

Instrumentation: Ansys HFSS, radio receiver chains (RTL-SDR + LNA), antenna design

INDUSTRY RESEARCH EXPERIENCE

Position Imaging

Stratham, NH

Machine Learning Engineer

May 2026 – Present

- Developed and productionized **graph neural networks** for RFID localization in multipath-dominated environments, owning the full stack from model architecture and training through GPU optimization, edge deployment, and production monitoring.
- Designed training systems using **reinforcement learning** and **neural surrogate models** to optimize computationally expensive GNN workloads and make large sweeps over architectures and training strategies affordable.
- Wrote custom **CUDA kernels** and GPU-level optimizations for performance-critical components of GNN training and inference; built edge inference pipelines covering model compilation, graph/operator optimization, hardware-specific acceleration, and integration with custom inference runtimes.
- Built the group's internal **ML experimentation platform** (ClearML, Azure, Datadog) for reproducible training, model evaluation, and rapid iteration, sustaining **30+ experiments/day**; added production observability for model behaviour, inference performance, and system health.

Thespian Labs

Somerville, MA

AI Engineer

Nov 2025 – Feb 2026

- Developed state-of-the-art **Text2Motion** foundation models using **VQ-VAE** motion tokenization; ran multimodal pre-training on 5000+ hours of human-performance time-series and DARTControl-style **reinforcement-learning post-training** for controllable motion synthesis.
- Released the open-source MLOps library **jobber** for programmatic research-job submission to cloud platforms.

Absentia Technologies

Boston, MA

Founding Machine Learning Engineer

Jan 2025 – Nov 2025

- Developed state-of-the-art **vision-language models (VLMs)** for multimodal video understanding; combined large-scale pre-training with **supervised fine-tuning and alignment post-training** on domain-specific data.

- Implemented distributed training pipelines with **PyTorch FSDP** (multi-GPU, mixed-precision) to train foundation models beyond single-device memory, handling sharding, checkpointing, and fault recovery.

Halo AI (Columbia-incubated stealth)

New York, NY (Remote)

Founding AI Engineer

Dec 2023 – Aug 2024

- Researched on-device **federated** and **ensemble** LLMs for agentic assistants under tight compute and memory constraints, including ablations on federated-aggregation strategies.
- Systematic study of **model-compression** techniques — pruning, knowledge distillation, and 8-bit post-training quantization — and their trade-offs for latency, memory footprint, and task accuracy (50% size reduction, 80% inference speedup, 90% accuracy retention).

Spatialise

Noordwijk, The Netherlands (Remote)

Geospatial Machine Learning Engineer

Feb 2025 – May 2025

- Developed a **multimodal spatiotemporal foundation model** over multispectral satellite imagery for soil-health remote sensing; combined deep learning with statistical priors (Gaussian-process regression, clustering) to produce calibrated predictions.

The KeelWorks Foundation

Oak Harbor, WA (Remote)

Software Engineer, ML Applications

Jul 2024 – Jan 2025

- Synthetic-data generation research via **Mistral-7B** fine-tuning (back-translation, sequential style representation); agentic LLM systems with RAG pipelines for large-scale document retrieval and question answering.

ASTRONOMY SCHOOLS AND WORKSHOPS

Inter-University Center for Astronomy and Astrophysics (IUCAA)

Pune, India

Introductory Summer School in Astronomy and Astrophysics

May – Jun 2020

REVIEW EXPERIENCE

- United States Research Software Engineer Conference (US-RSE) 2025, 2026